

INGI2142

Segment Routing

RSVP-TE

- What is the real scaling limit with RSVP-TE today ?

IGP costs

- How do network operators set IGP costs ?
- Is it possible to reach those objectives with a single cost per link ?
- Could we use several IGP costs per link ?

LDP-IGP synchronisation

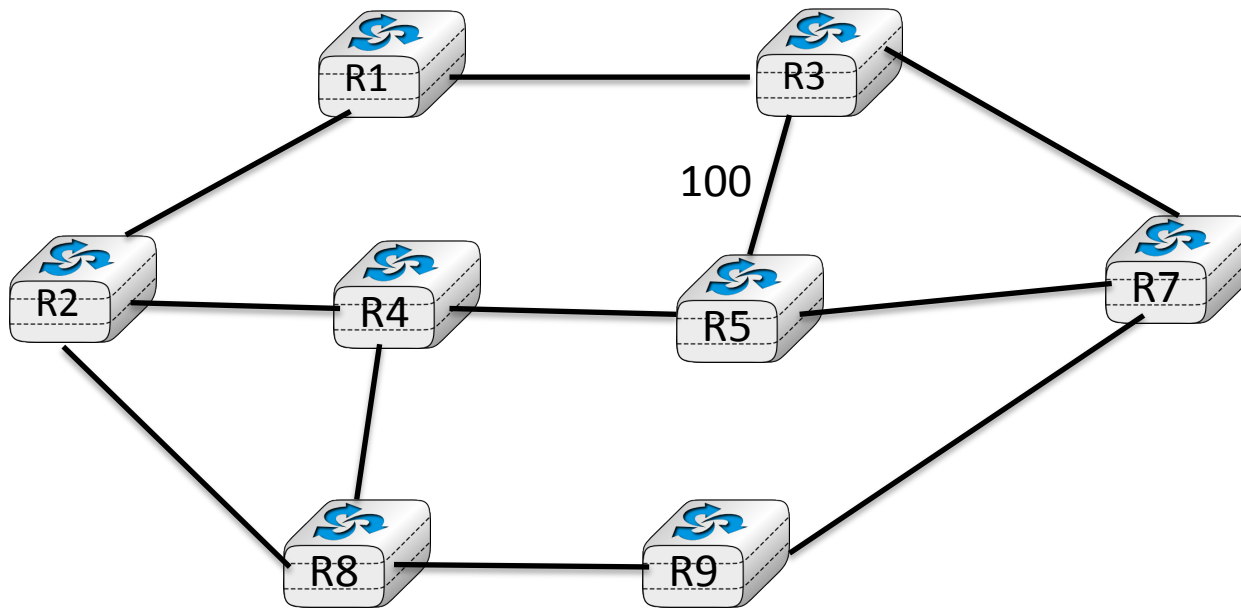
- Can some explain this problem with his/her own words and an example ?

Segment Routing

- How does source routing worked
 - In IPv4
 - In IPv6
- Why has the Segment Routing architecture been introduced ?

Node segments

- What is a node segment ?
- How are they advertised ?

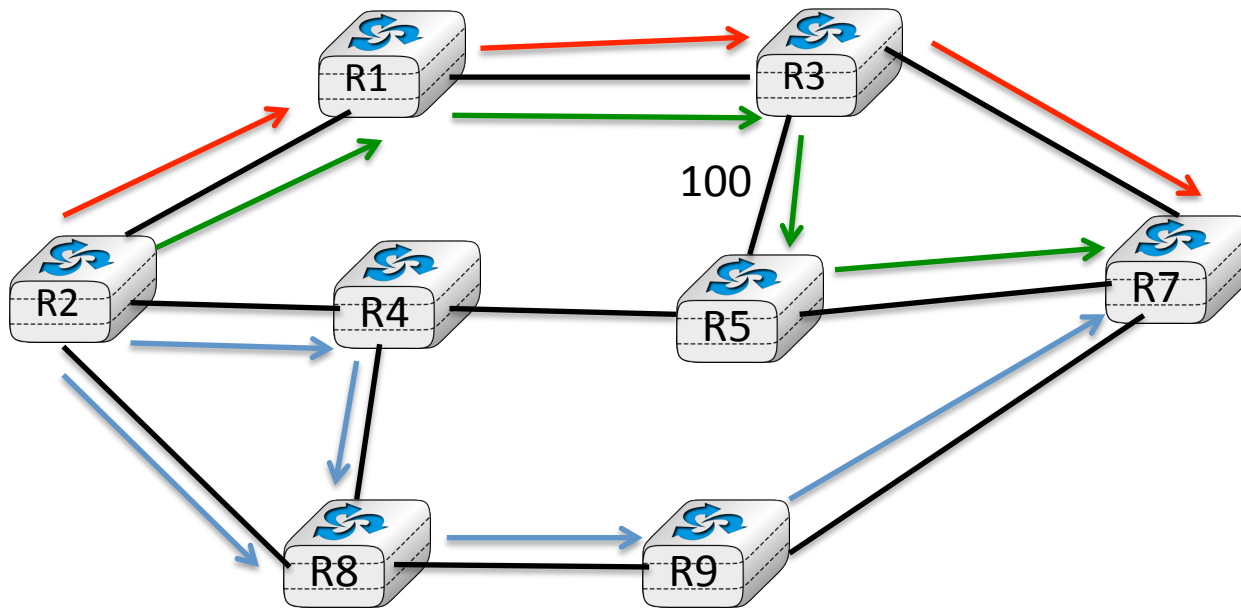


State

- Which state needs to be maintained
 - On the core routers
 - On the edge routers
- Who decides which path will be used for a particular traffic flow ?

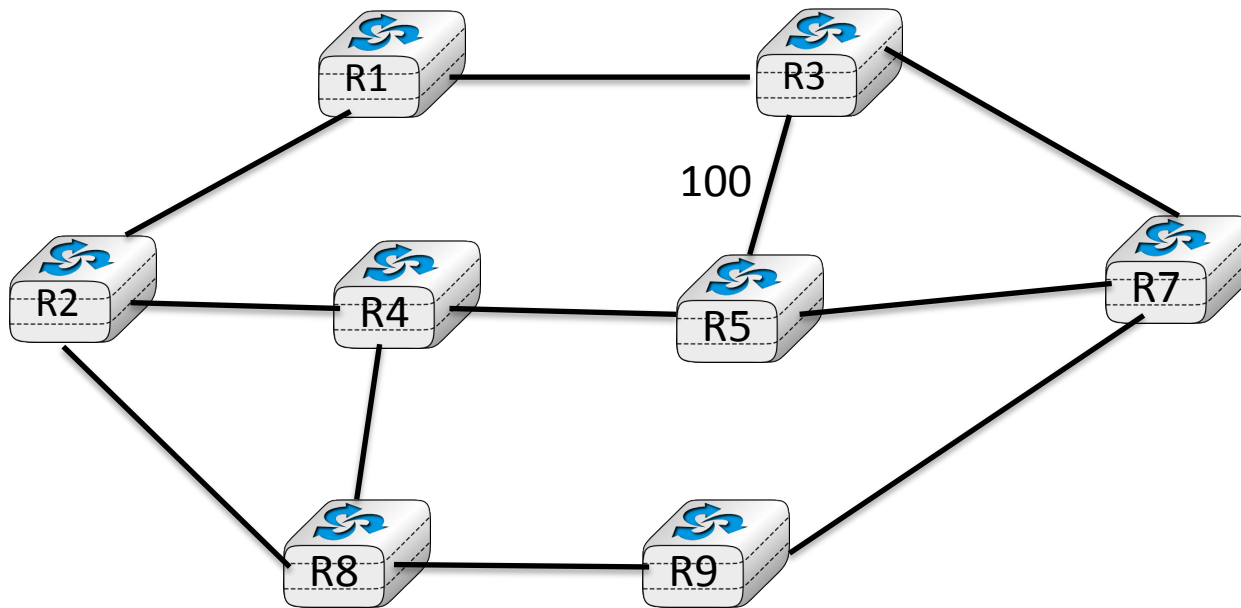
Paths

- Which paths can be realised by using node segments ?
 - How many labels do you need ?



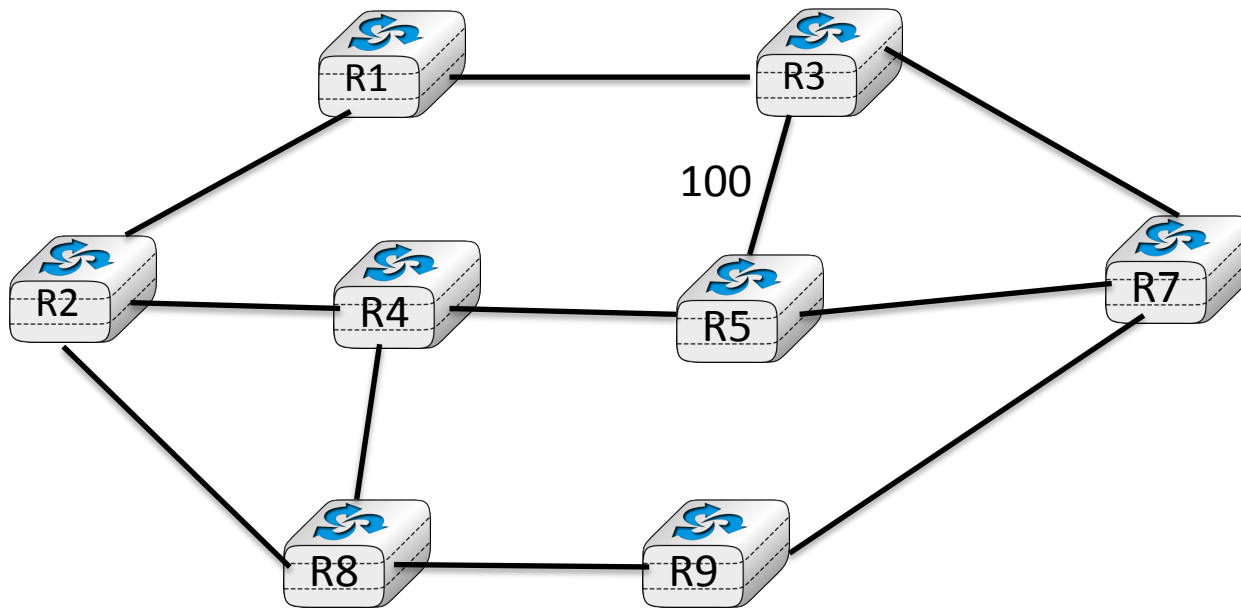
Node segments

- Can you show one path which can be realised by using RSVP-TE and not by using SR ?



Node segments

- Can you show one path which can be realised by using LDP and not by using SR ?

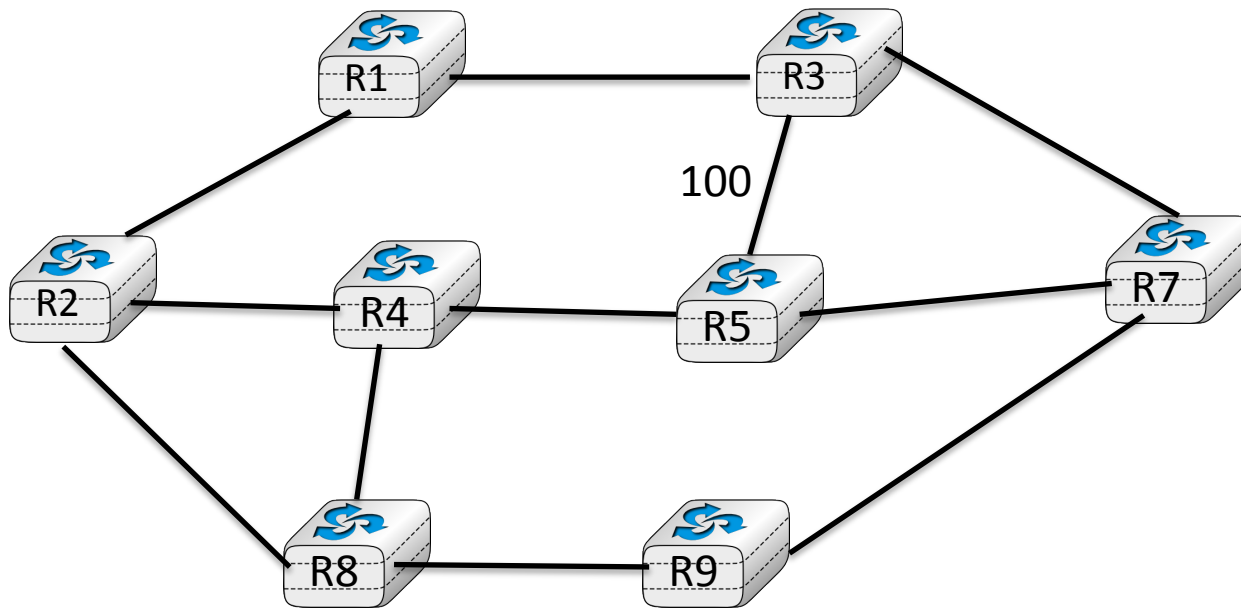


Limitations of the node segments

- What is the impact of the utilisation of node segments on the scalability of the link state routing protocol ?
- How many node segments are required to create specific paths ?
- Are there paths that cannot be realised by using only node segments ?

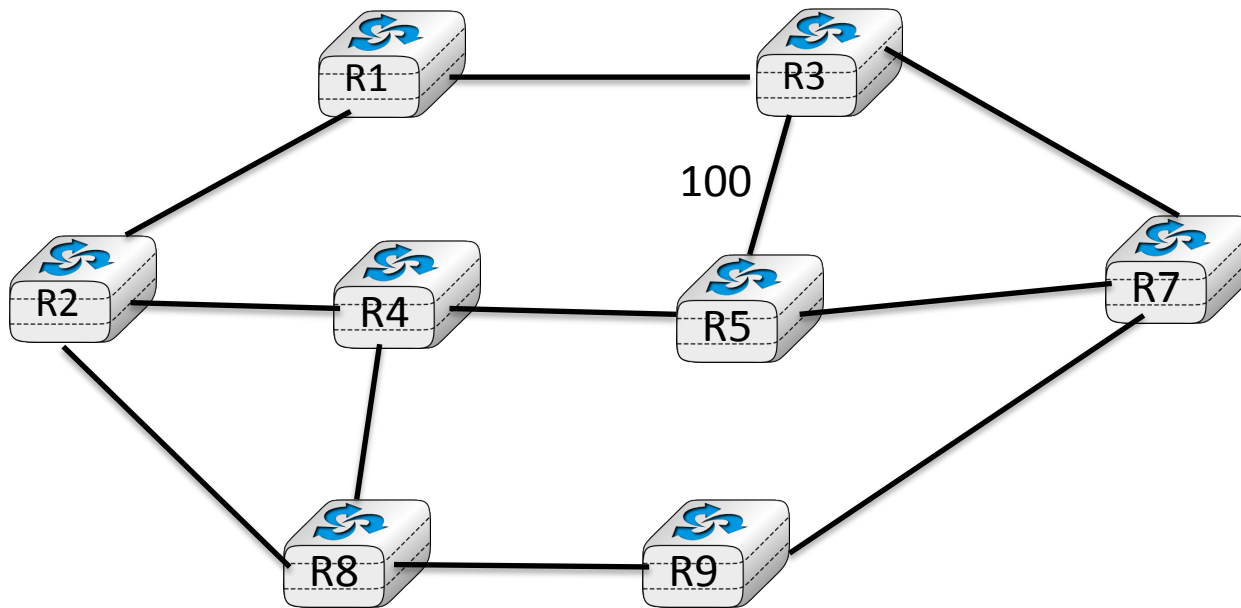
Adjacency segments

- What is an adjacency segment ?
- How are they advertised ?



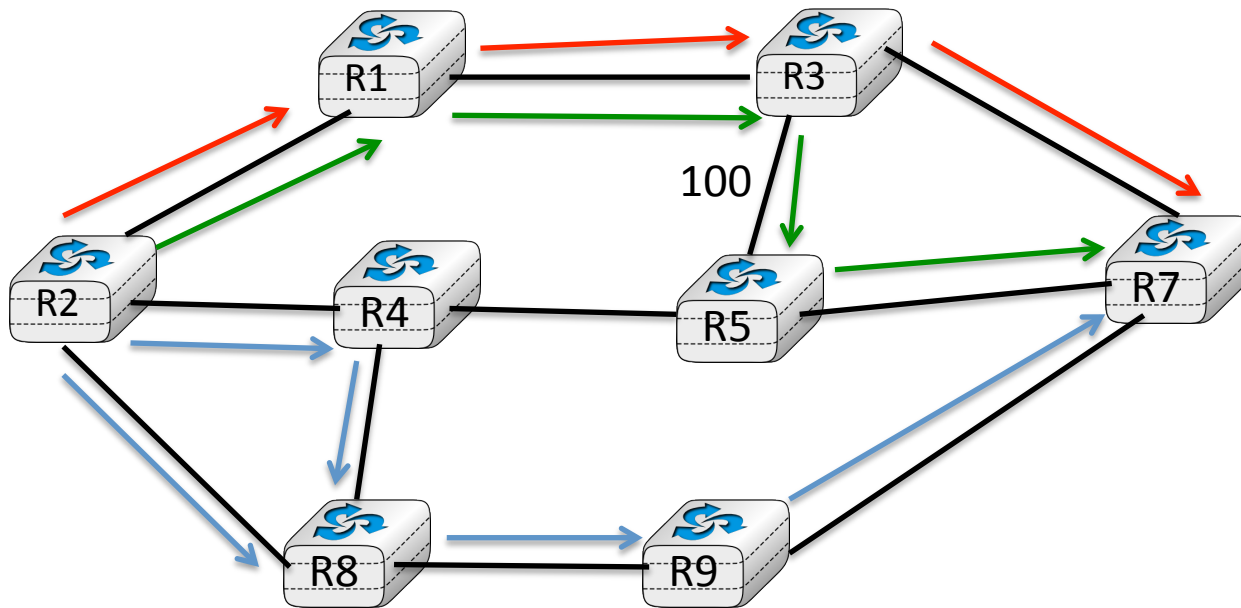
Adjacency segments

- Can you show one path that requires one adjacency segment ?



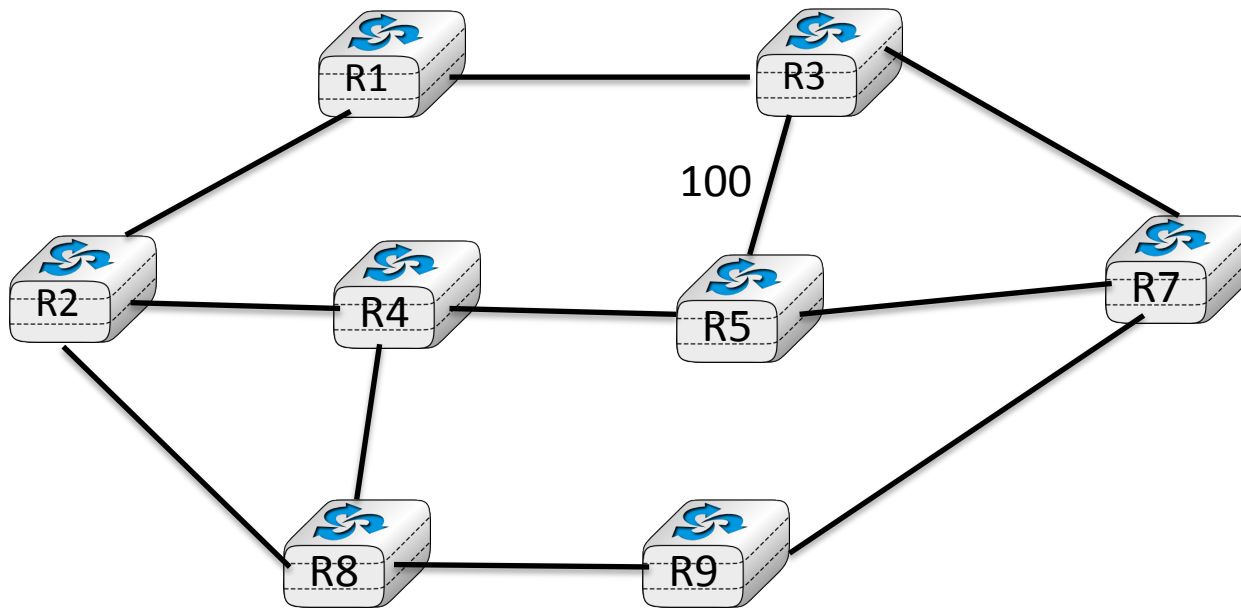
Paths

- Which paths require the utilisation of adjacency segments ?
 - How many labels do you need for each path ?



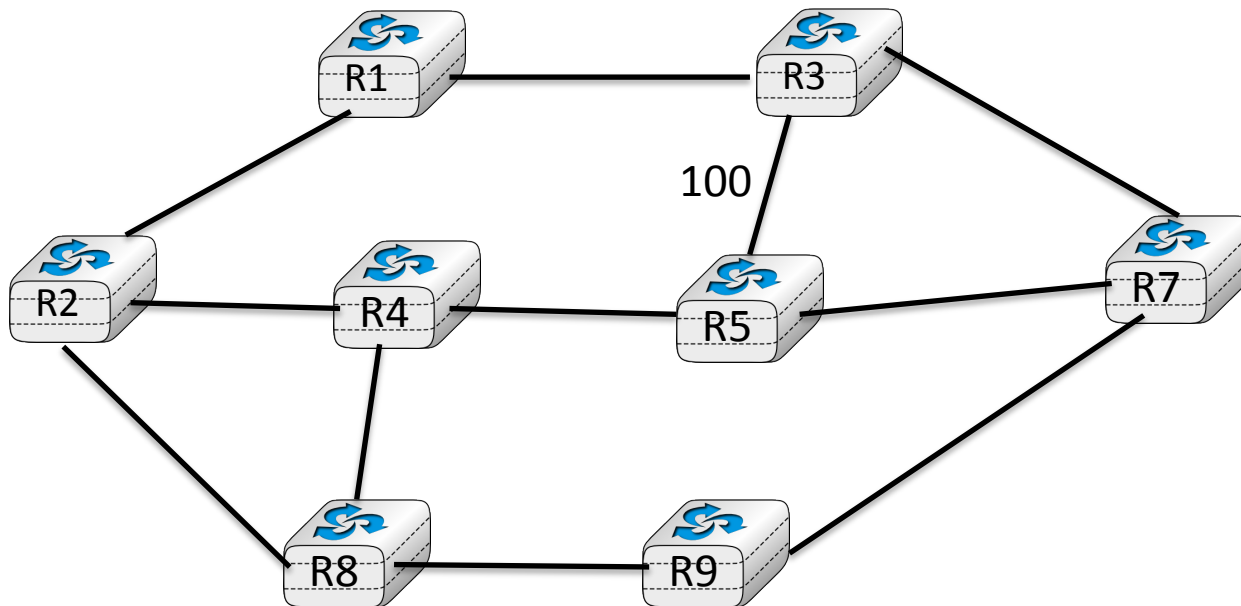
Adjacency segments

- Can you show one path which can be realised by using RSVP-TE and not by using SR with adjacency segments?



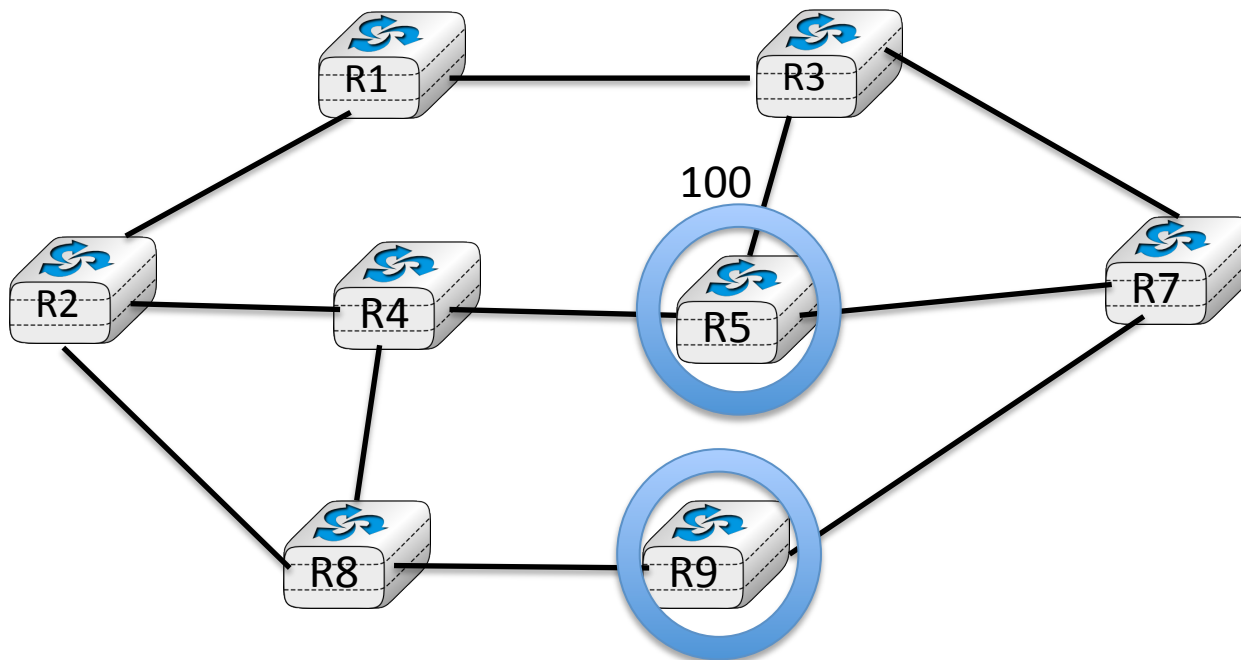
Adjacency segments

- Can you show one path which can be realised by using LDP and not by using SR with adjacency segments ?



Anycast segment

- What happens if multiple routers advertise the same node segment in the network ?

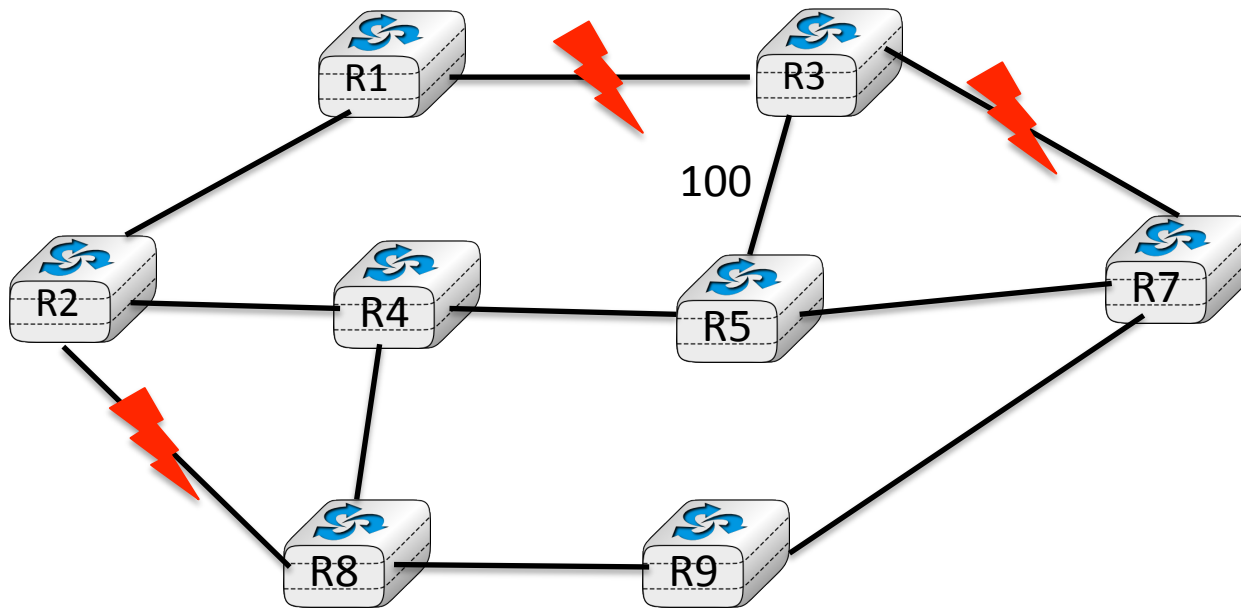


The dataplane

- How does SR operate with an MPLS dataplane ?
- How does SR operate with an IPv6 dataplane ?

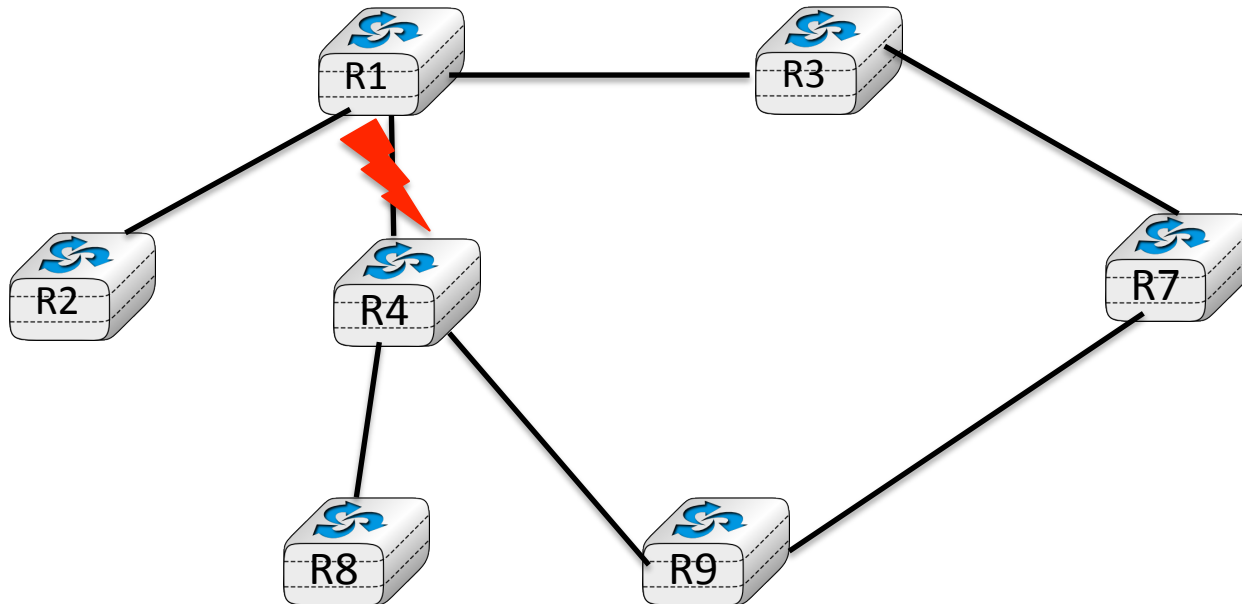
Use cases

- Failure protection
- Loop Free Alternate



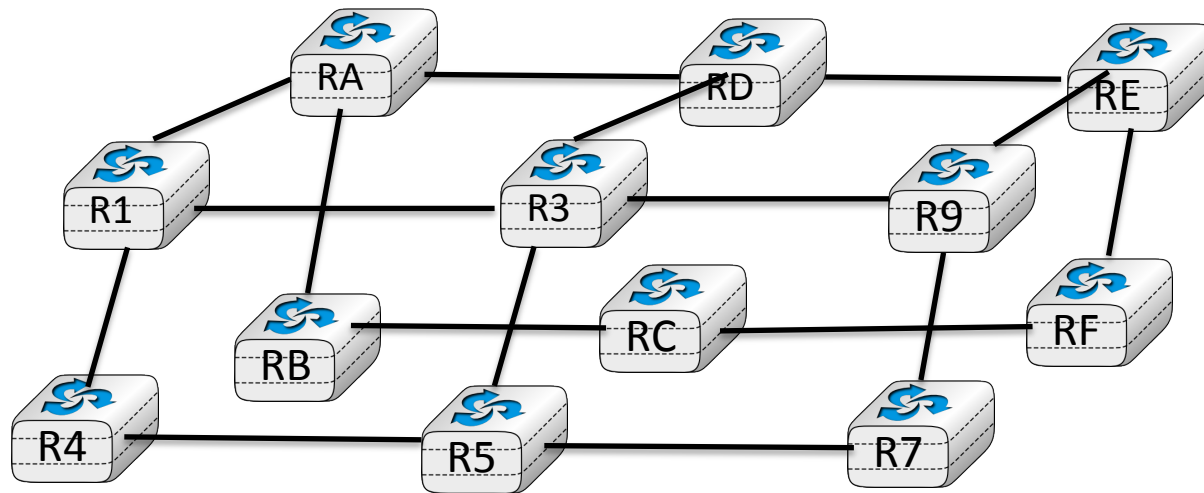
Remote LFA

- Usage of Remote LFA in this scenario



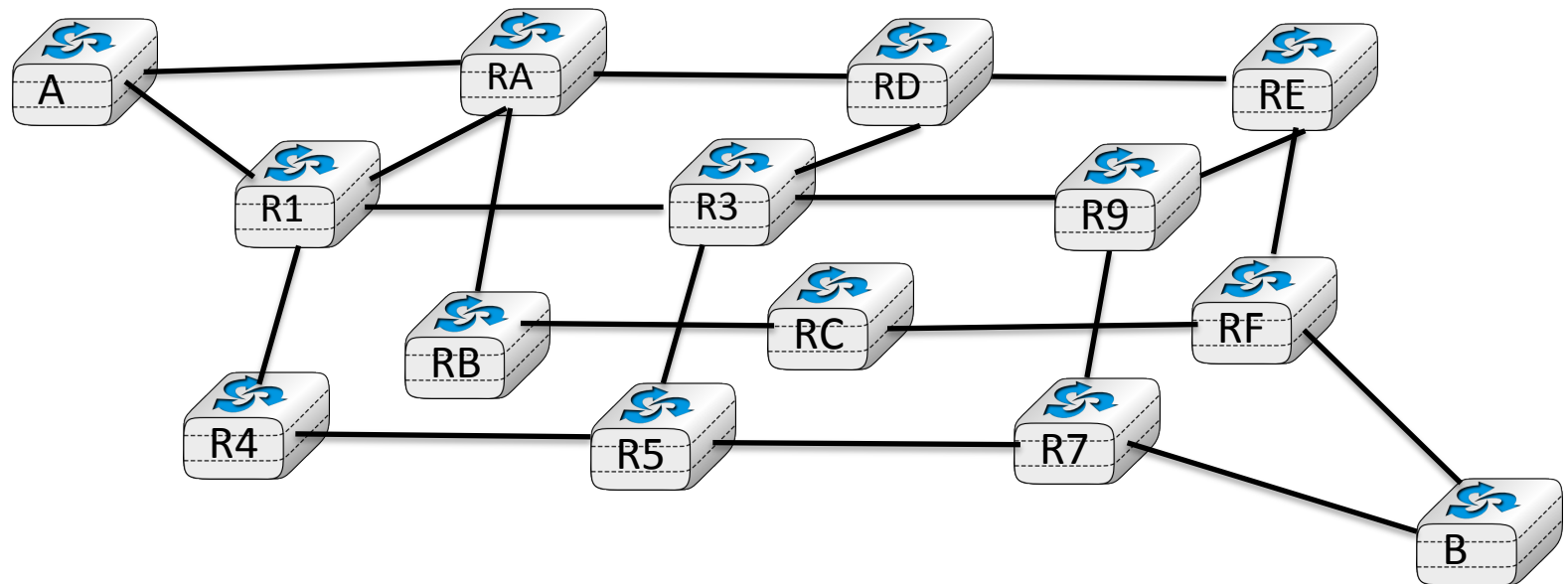
Dual-plane networks

- Frequent architecture in national networks



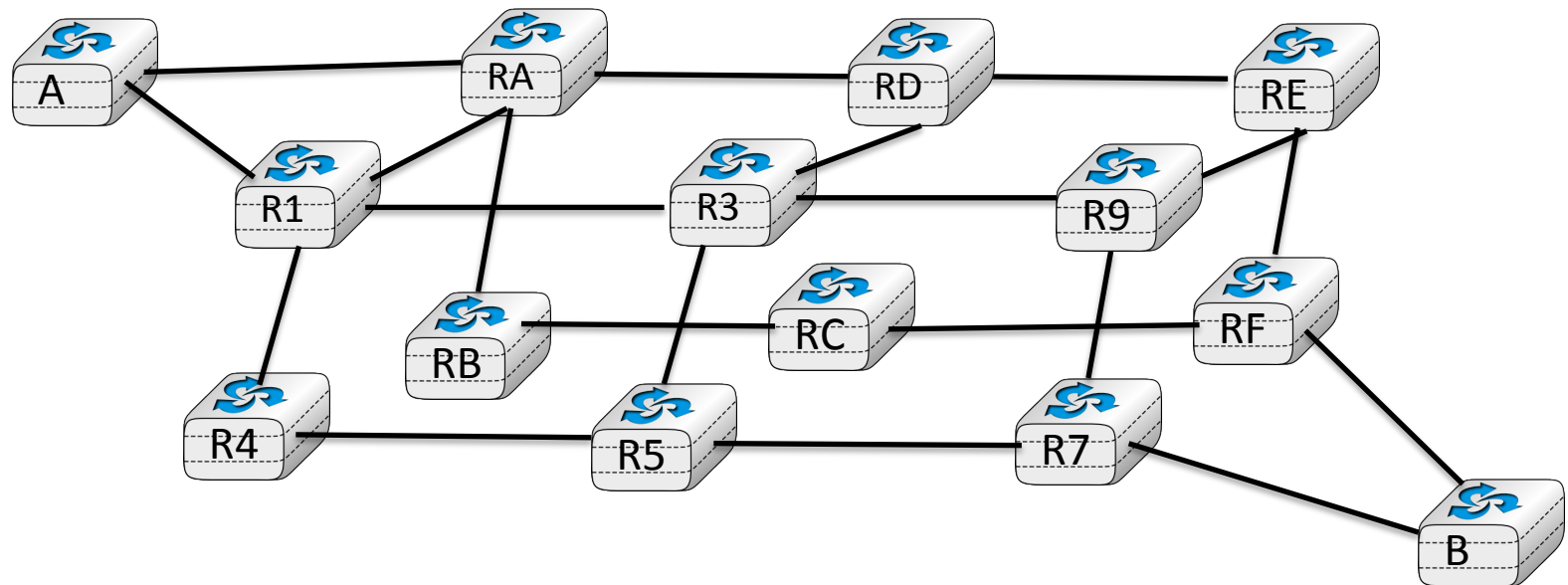
Load balancing

- How to send traffic on both planes ?



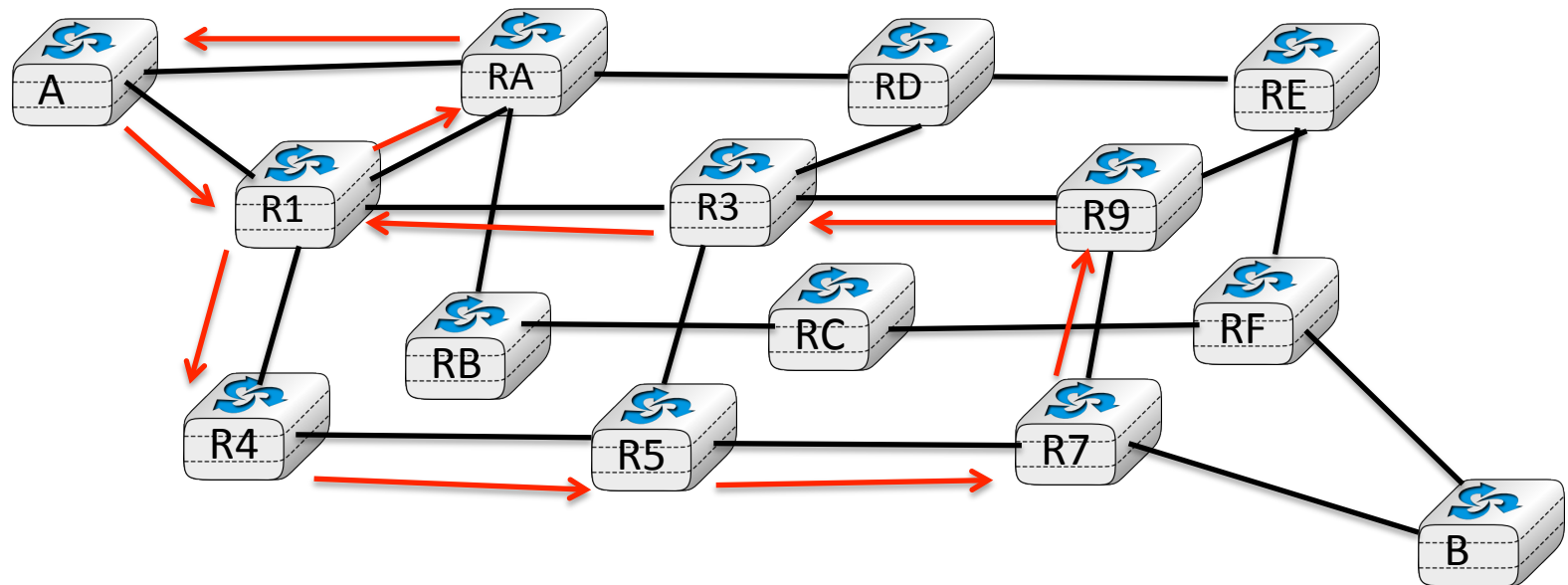
Disjoint paths

- How to send traffic over two disjoint paths ?



Crazy paths

- Is it possible to define paths that loop ?
 - Why would they be useful ?



Binding/service segments

- Can we use other segments than the node and adjacency segments ?
- What would be the utilisation of such segments ?